



Point, Line Segment, Line and Ray

I. POINT

A dot ($\bullet$) represents a point.

We name a point by a capital letter A, B, P, Q etc.

Mark a dot with the help of a fine pencil on a piece of paper. Name it A.

We say that A is a point.

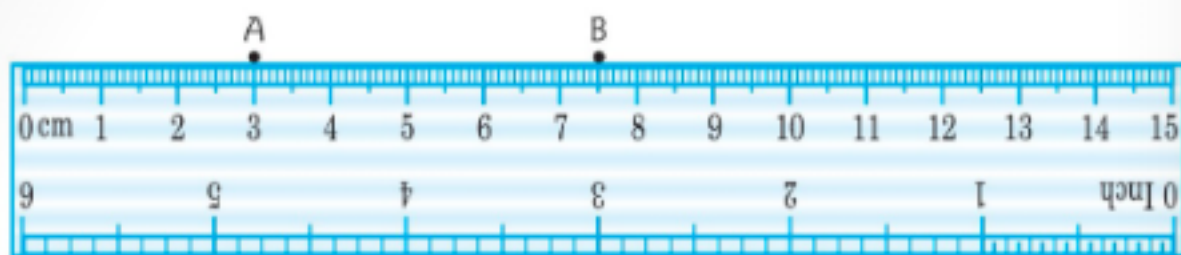
$\bullet$ A

A point shows a definite position. A point has no length, breadth or thickness.

II. LINE SEGMENT

Let us mark two points A and B on a paper.

We put a ruler in such a way that one of its straight edges touches both the points as shown below.



We hold the ruler firmly and move the pencil from A to B along the edge of the ruler.

We get a figure as shown on the right.

A ————— B

This is called a line segment AB, written as $\overline{AB}$.

Actually, the straight path from A to B is called the line segment $\overline{AB}$.

The points A and B are called the end points of $\overline{AB}$.

The distance between the points A and B is called the length of $\overline{AB}$.

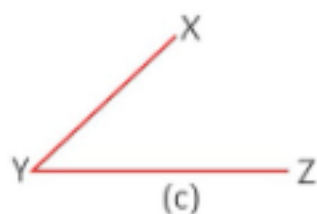
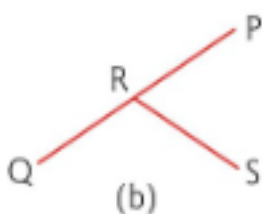
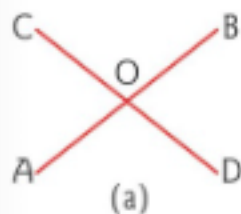
Properties of a Line Segment

- A line segment has two end points.
- A line segment has a definite length, but no breadth or thickness.
- A line segment can be drawn on a paper.

Whenever two line segments meet, they meet at a point.

The point at which two or more line segments meet is known as the point of intersection.





In figure (a), the line segments AB and CD meet at O . So, O is the point of intersection of line segments AB and CD .

In figure (b), the line segments PQ and RS meet at R . So, R is the point of intersection of line segments PQ and RS .

In figure (c), the line segments XY and YZ meet at Y . So, Y is the point of intersection of line segments XY and YZ .

III. LINE

A line segment AB extended endlessly on both sides is called a line and we denote it by $\overleftrightarrow{AB}$.

Actually, we cannot draw a line.

By drawing a line, we shall mean to draw a line segment AB , extend it in both ways and put arrow heads on both sides as shown here.



Properties of a Line

- A line has no end points.
- Since a line extends indefinitely in both the directions, it has no definite length.
- We represent a line AB by $\overleftrightarrow{AB}$.
- Practically, we cannot draw a line on paper. We can only draw a part of a line and put arrow heads on both the sides.



IV. RAY

A line segment extended endlessly in one direction is called a ray.

Thus, a line segment AB extending endlessly in the direction from A to B and marked by an arrow mark at B , represents a ray $\overrightarrow{AB}$.

Properties of a Ray

- A ray $\overrightarrow{AB}$ has one end point, namely A .
This point A is called the initial point of ray $\overrightarrow{AB}$.
- A ray has no definite length.
- Since a ray is endless in one direction, it cannot be drawn on a paper.

By drawing a ray $\overrightarrow{AB}$, we would mean that we draw a line segment AB , extend it from B onwards and put an arrow head as shown.



Different Types of Lines

(a) Horizontal Lines (Sleeping Lines):

In the given figure, a pencil is lying flat on the ground.

We say that the pencil is lying horizontally.

Similarly, a line drawn in a sleeping position is called a horizontal line.

Thus, in the adjoining figure, $\overleftrightarrow{AB}$ is a horizontal line.



(b) Vertical Lines (Standing Lines):

Look at a flagmast standing on the ground or an electric pole on the roadside. We say that the flagmast as well as the electric pole are standing vertically on the ground.

Similarly, a line drawn in a standing position is called a vertical line.

Thus, in the adjoining figure, $\overleftrightarrow{CD}$ is a vertical line.

Anything 'lying flat' is said to be in a horizontal position, while anything 'standing straight upright' is said to be in a vertical position.

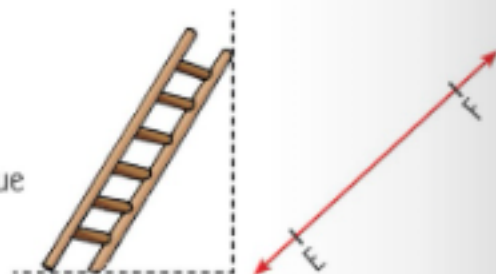


(c) Oblique or Slanting Lines:

Look at the figure of a ladder on a vertical wall. The ladder is in a slanting position. Similarly, a line drawn in the slanting position is called an oblique line.

Thus, in the adjoining figure, $\overleftrightarrow{EF}$ is an oblique line.

A line which is neither horizontal nor vertical, is called an oblique or a slanting line.



Exercise 62

1. Name the points shown in the figure.

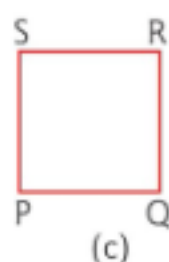
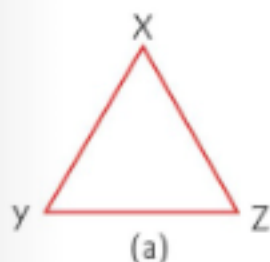
P •

Q •

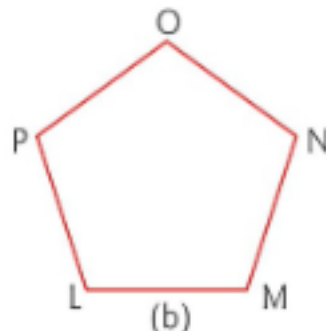
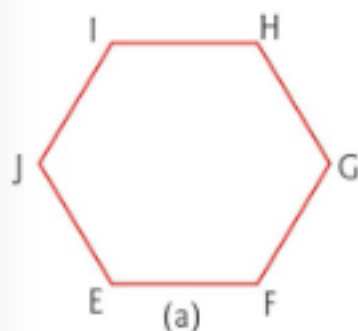
2. Name the points marked between P and Q on the line segment $\overline{PQ}$.



3. Name the line segments in each of the following figures.



4. Name the line segments in each of the figures given below.



5. Mark a point P on your paper. Draw six line segments, each containing the marked point.

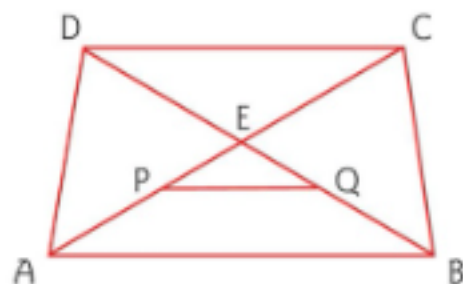
Can you draw some more line segments?

6. Mark two points M and N on a sheet of paper as shown: M • N •

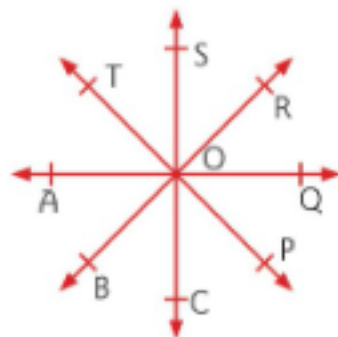
How many line segments can you draw passing through both these points?

7. Look at the figure and answer the following questions.

- Where do AC and BD meet?
- Where do PQ and AE meet?
- Where do PC and QD meet?
- At what point do AD and CD meet?
- Name seven points in the given figure.
- Name seven line segments in the given figure.



8. Write the names of the rays from the figure given below:



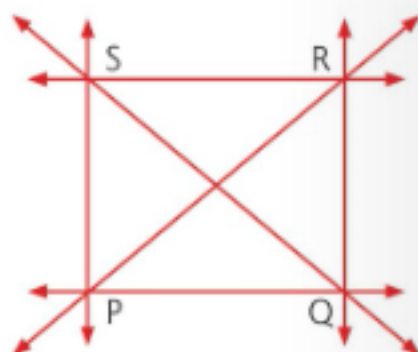
9. Fill in the blanks.

- (a) A line segment has end points.
- (b) A line has end points.
- (c) A ray has end point.
- (d) A line segment has a length.
- (e) A line AB is represented by
- (f) A ray AB is represented by
- (g) A dot ($\bullet$) represents a
- (h) A point shows a definite



10. In the given figure,

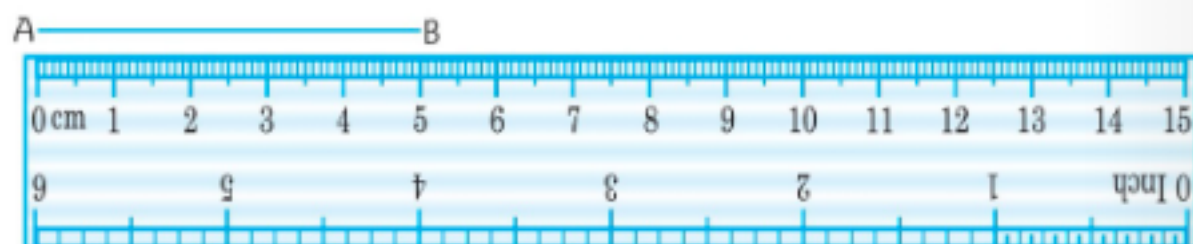
- (a) name the horizontal lines.
- (b) name the vertical lines.
- (c) name the oblique lines.



Measuring Line Segments

As we have learnt earlier, we measure the length of a line segment with the help of a ruler or a 15 cm scale.

Suppose we have to measure the length of a line segment $\overline{AB}$ as shown below. Put the ruler along the line in such a way that one of its edges touches both the points A and B and the 0 mark of the ruler is at A.



Now, read the ruler mark at B.

The reading of the scale at B gives the length of the line segment $\overline{AB}$ in cm.

Here, it reads 5 cm.

Therefore, length of $\overline{AB} = 5$ cm.

To Draw a Line Segment of a Given Length

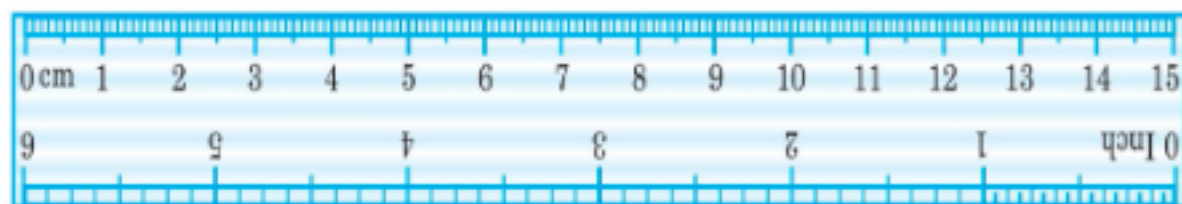
Suppose we have to draw a line segment of length 6 cm. We proceed as follows:

Step 1: Put the ruler on a piece of paper. Hold it firmly.

Step 2: With the help of a sharp pencil, mark a point P against the 0 mark of the ruler and a point Q against the 6 cm mark of the ruler.

Step 3: Move the pencil from P to Q along the edge of the ruler.

P ————— Q



Thus, we obtain the line segment PQ whose measure is 6 cm.



Exercise 63

1. Construct a line segment AB of length 7 cm.
2. Construct a line segment XY of length 4 cm.
3. Construct a line segment of length 8 cm.
4. Construct a line segment of length 9 cm.
5. Measure each of the following line segments.



(a) $\overline{AB} = \dots\dots$ cm



(b) $\overline{CD} = \dots\dots$ cm



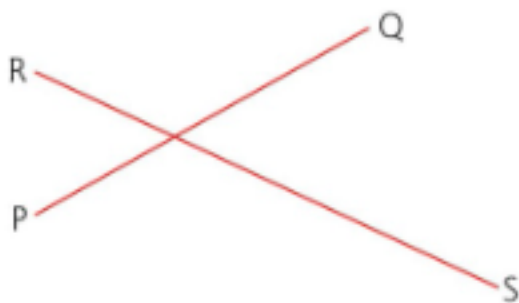
(c) $\overline{EF} = \dots\dots$ cm



(d) $\overline{GH} = \dots\dots$ cm



(e) $\overline{PQ} = \dots\dots$ cm and $\overline{RS} = \dots\dots$ cm



6. Measure each side of the given figure and write in the space provided.



AB =

BC =

CD =

DA =

Triangle, Quadrilateral, Rectangle, Square and Circle

Triangle

A figure bounded by three line segments is called a **triangle**.

The three line segments forming a triangle are called its sides.

The point at which two sides of a triangle meet is called a vertex of the triangle.

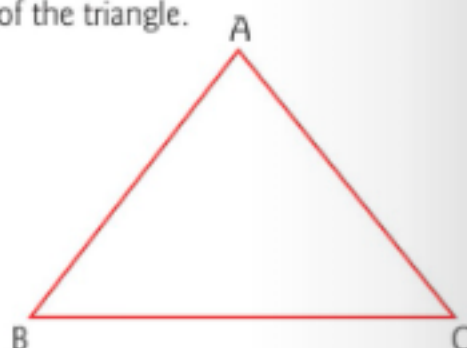
A triangle has 3 sides and 3 vertices (or corners).

We name a triangle by its vertices.

In the given figure, ABC is a triangle.

This triangle has

- (a) three vertices, namely A, B and C.
- (b) three sides namely AB, BC and CA.



Quadrilateral

A figure bounded by four line segments is called a **quadrilateral**.

The four line segments forming a quadrilateral are called its sides.

The point at which two sides of a quadrilateral meet is called its vertex.

A quadrilateral has 4 sides and 4 vertices (or corners).

We name a quadrilateral by its vertices.

In the given figure, ABCD is a quadrilateral.

This quadrilateral has

- (a) four vertices, namely A, B, C and D.
- (b) four sides, namely AB, BC, CD and DA.



Rectangle

A quadrilateral bounded by two horizontal line segments and two vertical line segments is called a **rectangle**.

The opposite sides of a rectangle are equal. In the given figure,

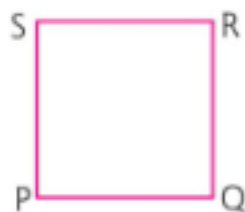
ABCD is a rectangle in which $AB = DC$ and $BC = AD$.



Square

A rectangle having all sides of the same length is called a square.

In the figure, PQRS is a square in which $\overline{PQ} = \overline{QR} = \overline{RS} = \overline{SP}$



Circle

If we place a bangle or a rupee coin on a piece of paper and move a pencil around it, we get a figure shown alongside.

We call it a circle.

It has no sides and no vertices.



Exercise 64

1. Name each of the following figures.

(a)



.....

(b)



.....

(c)



.....

(d)



.....

(e)



.....



2. Which shape do the following have?

(a) The wheel of a bicycle

(b) A blackboard

(c) A page of the book

(d) The face of the full moon

(e) A set-square in a geometry box

(f) A postcard

(g) The surface of a book

(h) The face of a dice

3. Fill in the blanks.

(a) A triangle has sides and vertices.

(b) A rectangle has sides and vertices.

(c) All the sides of a square are

(d) The sides of a rectangle are equal.

(e) A circle has side and vertex.

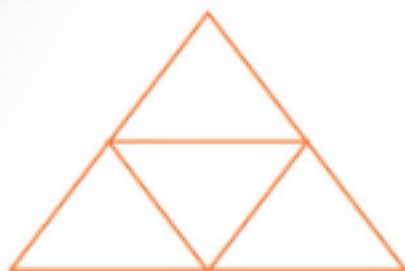
4. Look at the following figures and count the number of triangles in each case.

(a)



.....

(b)



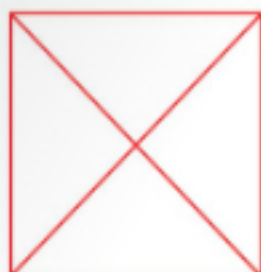
.....

(c)



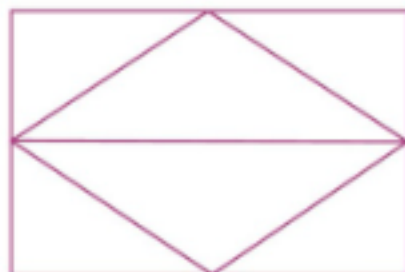
.....

(d)



.....

(e)



.....

(f)



.....

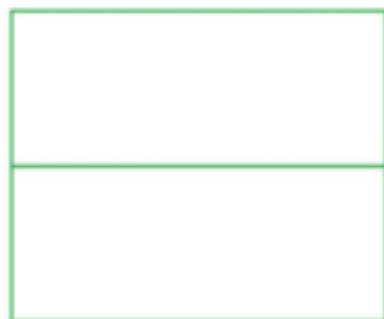
(g)



.....

5. Look at the following figures and count the number of rectangles in each case.

(a)



.....

(b)



.....

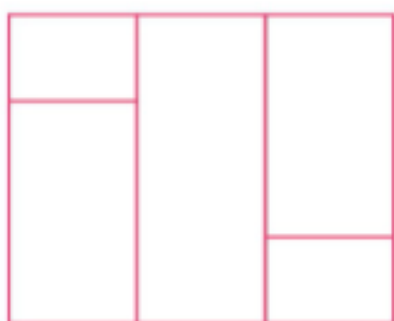
(c)



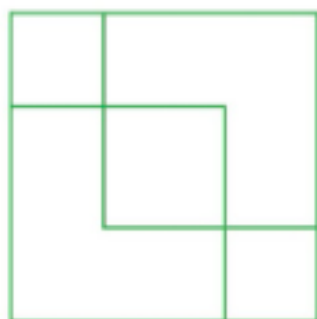
.....



6. Count the number of rectangles in the following figure:



7. Count the number of squares in the following figure:



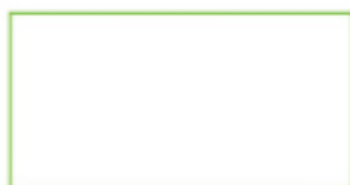
Can You Do It

1. (a) Can you draw 2 lines to split the shape into 3 triangles?



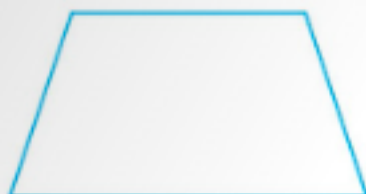
- (b) Can you draw one line to split the given shape into 3 triangles?

- (c) Can you draw 2 lines to split the shape into 3 triangles?



- (d) Can you draw 2 lines to split the shape into 4 triangles?

- (e) Can you draw 2 lines to split the shape into 3 triangles?



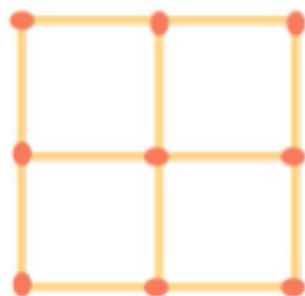
- (f) Can you draw 3 lines to split the given shape into 4 triangles?

- (g) Can you draw 3 lines to split the shape into 5 triangles?



- (h) Can you draw 3 lines to split the shape into 4 triangles?

2. Observe the following figure drawn with the help of matchsticks:



- (a) How many squares can you count in the above figure?
- (b) How many rectangles can you count in the above figure?
- (c) Can you remove two matchsticks so that the new figure has 2 squares only?
- (d) Can you remove two matchsticks so that the new figure has 3 squares only?
- (e) Can you remove two matchsticks so that the new figure has 2 rectangles inside a square?

Solids

Idea of Space

When we kick a ball, it goes up in the air. We say that the ball is moving in space. Similarly, we say that an aeroplane moves in space.



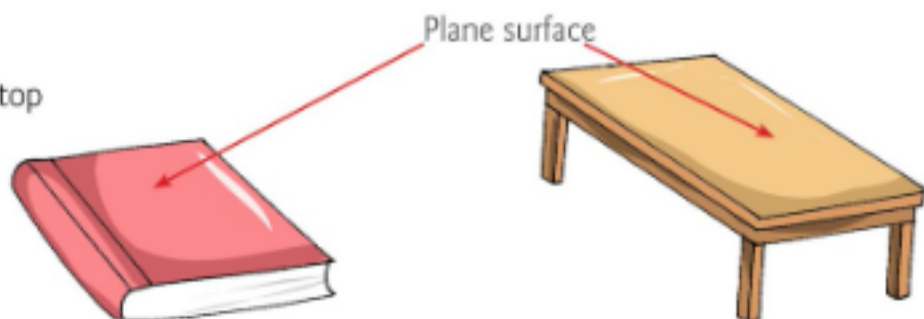
Solid

An object that occupies space is called a solid. Solids can be of various shapes.

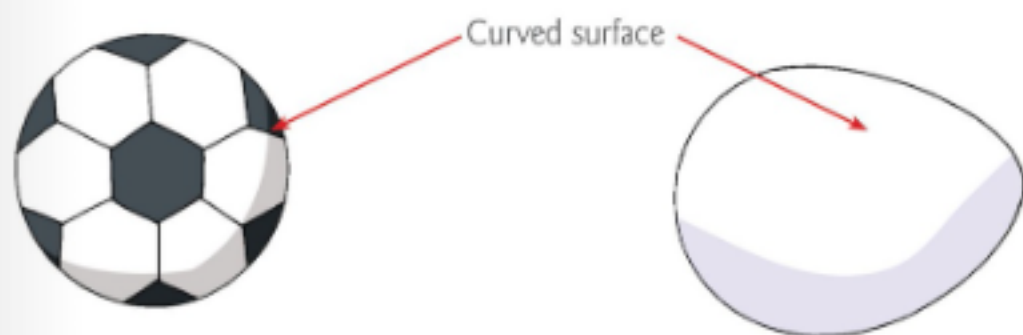
The outside of a solid is called its surface. The surface may be plane or curved.

A plane is a flat surface.

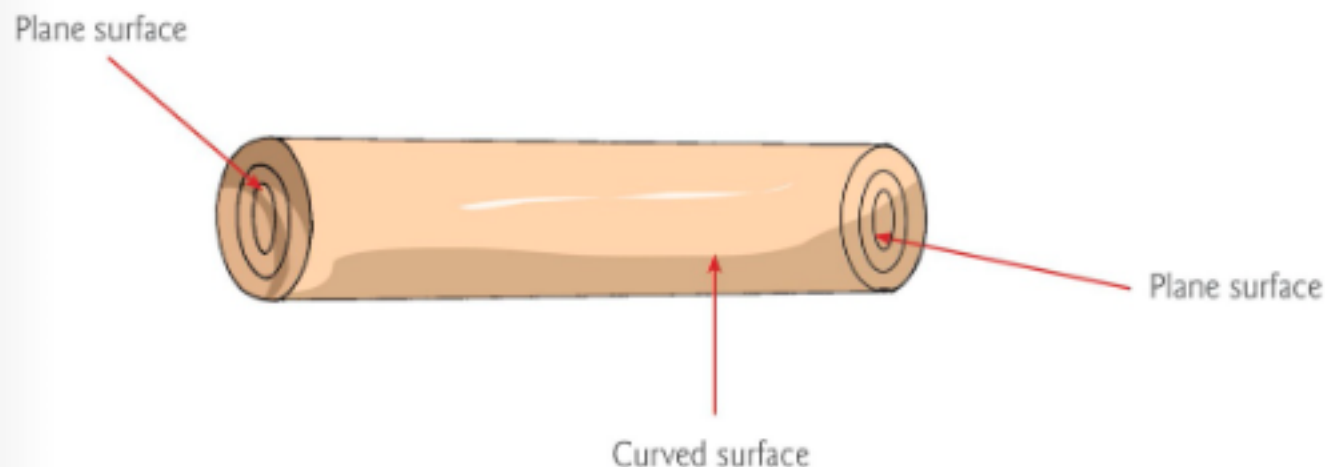
The floor of a room, the top of a desk, the surface of a blackboard are all plane surfaces.



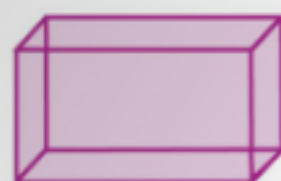
On the other hand, the surface of a football, an egg or a globe all have curved surfaces.



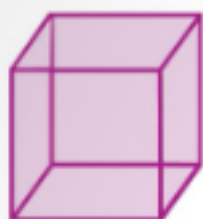
However, if you observe a log of wood, it has one curved surface and two plane surfaces.



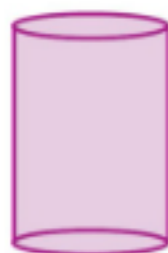
Some of the shapes of solids are:



(a) Cuboid



(b) Cube



(c) Cylinder



(d) Sphere



(e) Cone

Cuboid

Each of the solids — a wooden box, a match box, a chalk box, a tea-pack, a brick, a book, an almirah, etc. — is in the shape of a cuboid.



Suitcase



Match box



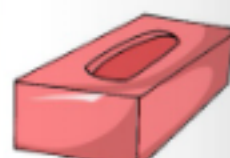
Tea-pack



Book



Almirah



Brick

Look at the cuboid shown in the adjoining figure.

A cuboid has six faces, each of which is a rectangle in shape.

In the figure, the six faces shown are ABCD, EFGH, DCGH, ABFE, BFGC and AEHD.

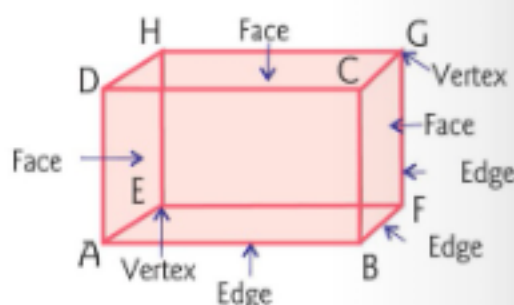
The opposite faces of a cuboid are identical.

Two adjacent faces of a cuboid meet at a line segment, which is called an edge of the cuboid.

A cuboid has 12 edges, namely AB, BF, EF, AE, AD, BC, FG, EH, DC, CG, GH, DH.

Three edges of a cuboid meet at a point, called a vertex.

A cuboid has 8 vertices, namely A, B, C, D, E, F, G and H.



Cube

A cuboid which has 6 identical faces is called a **cube**.

Thus, each face of a cube is a square.

Ice-cubes, sugar-cubes, dice etc. are all examples of a cube.



Ice-cube



Sugar-cubes

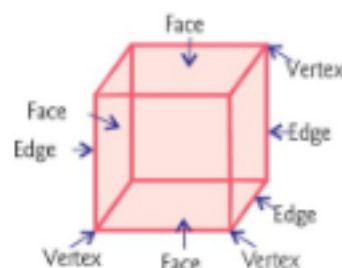


Chalk-box



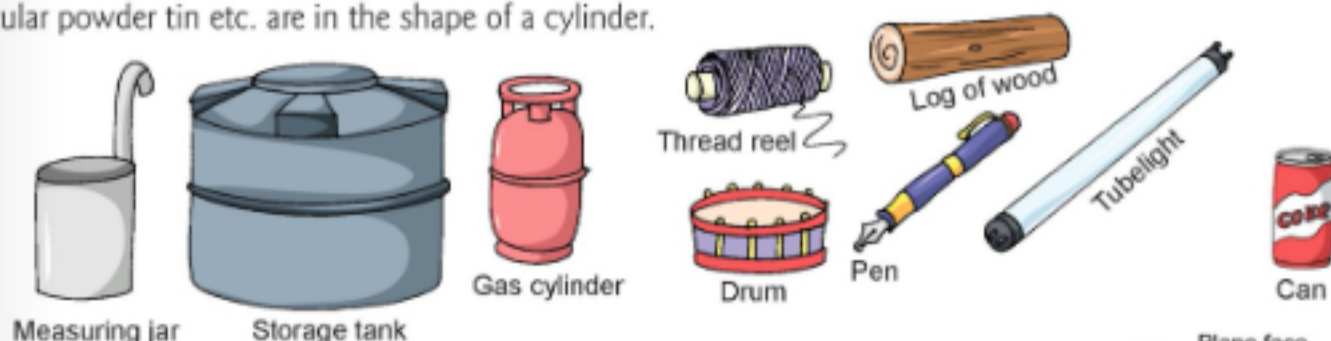
Dice

A cube has 6 faces, 12 edges and 8 vertices.



Cylinder

Objects such as a circular pillar, a circular pipe, a circular storage tank, a measuring jar, a gas cylinder, circular powder tin etc. are in the shape of a cylinder.

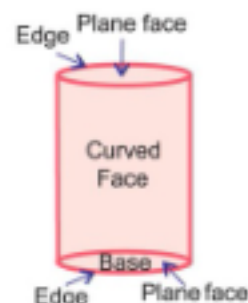


A cylinder has two plane faces and one curved face.

A cylinder has two edges, both of which are circular.

A cylinder has no vertex.

The base and the top of a cylinder are of the same shape and size.



Sphere

The objects which are in the shape of a ball are known to have the shape of a sphere.

A sphere has only one curved face.

A sphere has no vertex and no edge.



Cone

Objects such as an ice cream cone, a funnel, a conical tent, a conical vessel, a clown's cap, tapered end of a pencil are in the shape of a cone.



A cone has one plane face, which is its base.

A cone has one curved face.

A cone has one vertex.

A cone has one circular edge, where the curved face meets the plane face.

Summary:

Solid	Vertices	Plane/Flat faces	Curved faces	Edges
Cuboid	8	6	0	12
Cube	8	6	0	12
Cylinder	0	2	1	2
Sphere	0	0	1	0
Cone	1	1	1	1



Exercise 65

1. Tick (✓) the correct answer.

- (a) A cricket ball has a plane/curved surface.
- (b) The top of a book has a plane/curved surface.
- (c) An orange has a plane/curved surface.
- (d) The bottom of a brick has a plane/curved surface.

2. Identify the shape of each of the following objects.

- | | |
|-------------------------|----------------------|
| (a) Funnel | (b) Watermelon |
| (c) Battery cell | (d) Tubelight |
| (e) Pickle jar | (f) Dice |
| (g) Garden roller | (h) Candle |
| (i) Orange | (j) Pencil-box |

3. Fill in the blanks.

- (a) A cuboid has faces, edges and vertices.
- (b) The faces of a cuboid are identical.
- (c) All the faces of a are identical.
- (d) A has no vertex and no edge.



- (e) A has one plane face, one curved face, one vertex and one edge.
- (f) A cylinder has plane faces and edges.
- (g) A has two edges and no vertex.

4. Count the number of cubes in each of the following stacks:

(a)



(b)



(c)



(d)



(e)



(f)

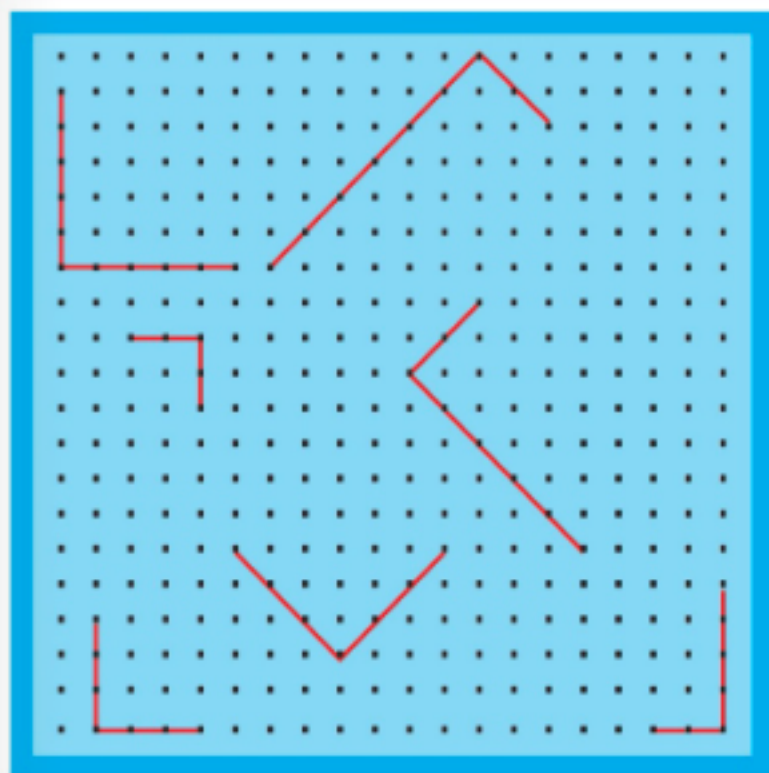


Activity Time



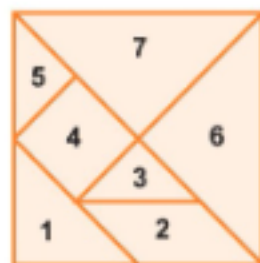
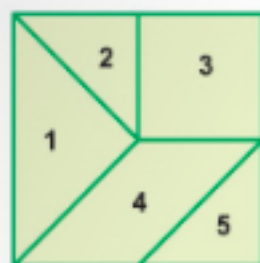
Activity 1. Dot-Grid

Shown below is a dot grid with some incomplete figures drawn on it. Complete these figures to make squares and rectangles by drawing just two lines in each figure.



Activity 2. Tangram

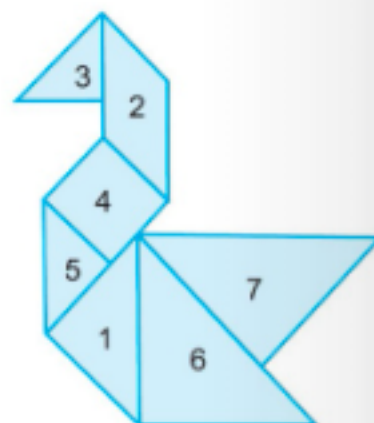
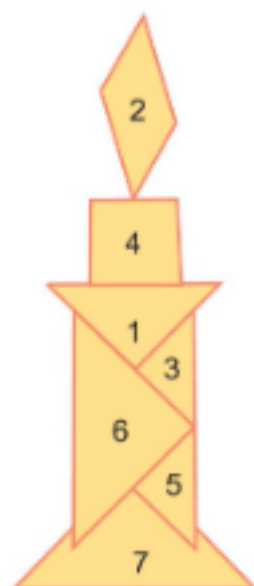
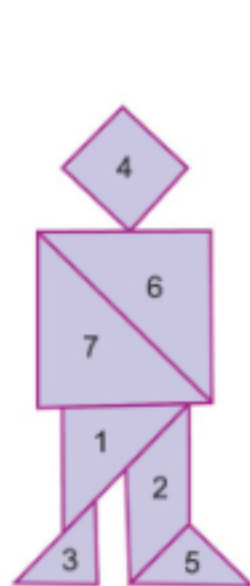
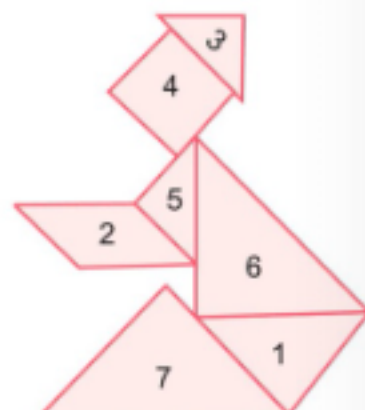
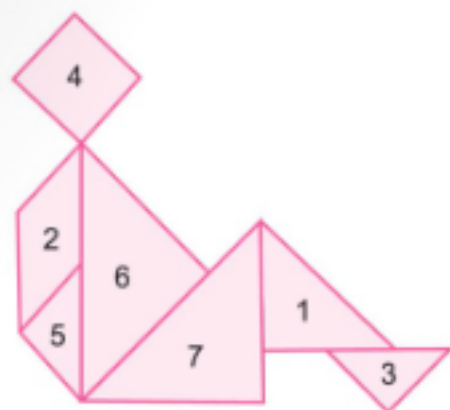
The tangram is an old Chinese puzzle, consisting of a square broken up into various geometrical shapes. There are two types of tangram—a 5-piece tangram and a 7-piece tangram. Both of them are shown below:



From the pieces of the tangram, we can make shapes of animals, people and things.

Two 7-piece tangrams have been given on the last page of the book. Cut out all the 7 pieces of one tangram separately. Place them on a hardboard and trace the outline of each piece with a pencil. Now, cut-out these shapes from the hardboard. Paste the cut-out pieces of the tangram on their respective hardboards. Now, cut out the pieces of the second tangram and paste them on the opposite sides of their respective hardboards.

Use these pieces to form different shapes as shown below:



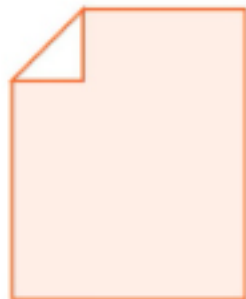
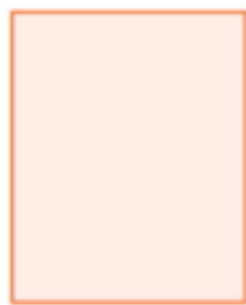
Try making some more similar shapes using this tangram.

1. Can you form a square using pieces 6 and 7?
2. Can you form a square using pieces 1, 2, 3, 4 and 5?
3. Can you form a rectangle using pieces 1, 2, 3 and 5?



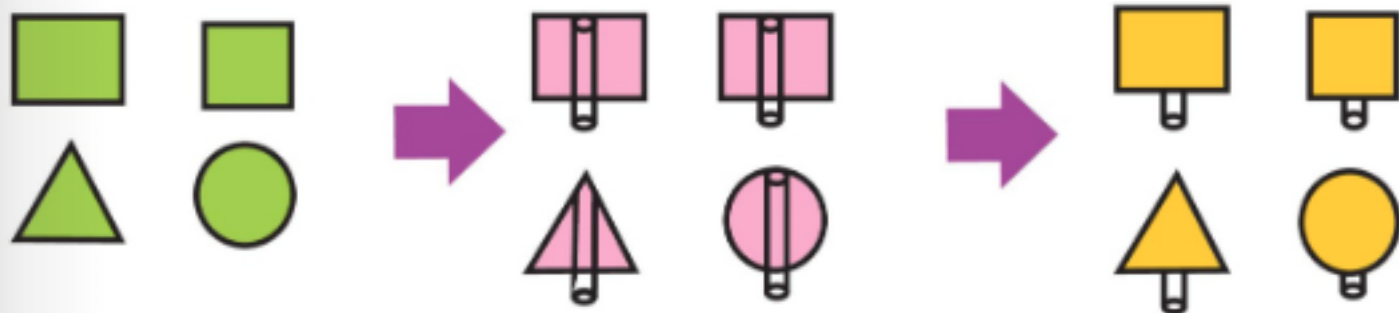
Activity 3. Edges and Corners

1. Take a rectangular sheet of paper.
How many corners and edges does it have?
2. Fold one of its corners.
How many corners and edges does it have now?
3. How many corners and edges will you get by folding two opposite corners of the sheet of paper?
4. Can you fold this paper in such a way that it has only three corners? You are allowed only two folds. What shape will you get?
5. Repeat the above activity with a square sheet of paper.
6. Can you fold all the corners of the square sheet in such a way that the number of corners remain unchanged?



Activity 4. (Class Activity)

Take a cardboard sheet and cut out from it the shapes of a square, a rectangle, a triangle and a circle. Now, fix a thin straw along the middle line on each of these shapes and then cover each of them with a glazed paper.



Mount each of these shapes on a battery-powered rotating motor and turn on the switch so that the shape rotates about the straw. What do you observe?

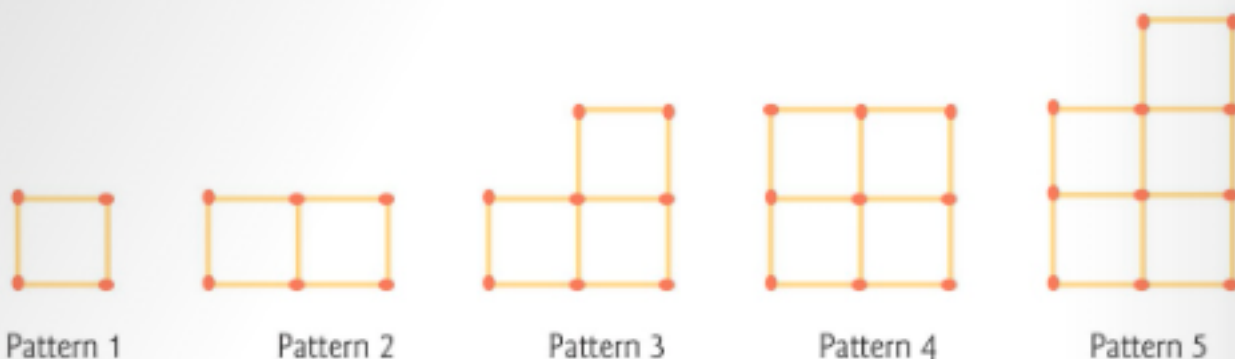
You will find that a rectangle and a square, on rotation generate a cylinder; a triangle generates a cone and a circle generates a sphere.

Activity 5. (Matchstick Patterns)

Use matchsticks or toothpicks to create shape patterns and count the number of matchsticks used and the number of shapes in the resulting pattern.

A few examples are shown below.

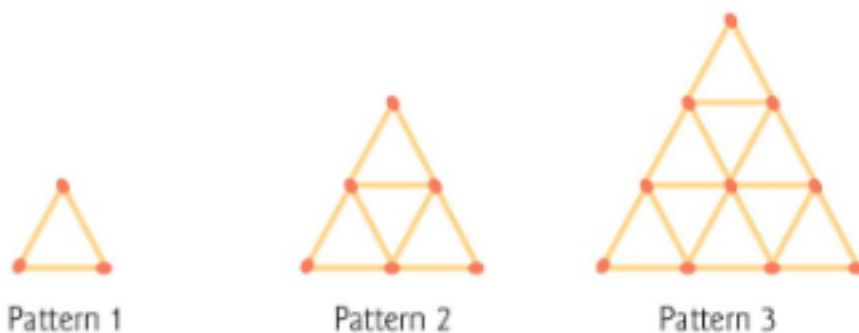
1.



Now, we have the following table:

Pattern No.	1	2	3	4	5
Number of matchsticks	4	7	10	12	15
Number of squares	1	2	3	5	6

2.



Now, complete the following table:

Pattern No.	1	2	3	4	5
Number of matchsticks					
Number of triangles					

Can you think of more such patterns?

**QUESTION BAG 1****(Objective Type Questions)****Tick (✓) the correct answer.****1.** Which of the following has no end point?

(a) Line

☐

(b) Line segment

☐

(c) Ray

☐

(d) None of these

☐**2.** Which of the following has a definite length?

(a) Line Segment

☐

(b) Ray

☐

(c) Line

☐

(d) Both (a) and (b)

☐**3.** A cuboid has vertices, faces and edges.

(a) 6, 8, 8

☐

(b) 8, 6, 12

☐

(c) 6, 6, 10

☐

(d) 6, 8, 12

☐**4.** Choose the odd one out.

(a) Rectangle

☐

(b) Triangle

☐

(c) Circle

☐

(d) Square

☐**5.** Which of the following statements is not correct?

(a) A line segment is a part of a line.

☐

(b) Every rectangle is a quadrilateral.

☐

(c) Every square is a rectangle.

☐

(d) A sphere has just one edge.

☐**6.** Which of the following solid shapes has no vertex?

(a) Cylinder

☐

(b) Sphere

☐

(c) Cone

☐

(d) Both (a) and (b)

☐**7.** Choose the incorrect statement.

(a) Line AB is the same as line BA.

☐

(b) Line segment AB is the same as line segment BA.

☐

(c) Ray AB is the same as ray BA.

☐

(d) All are correct.

☐**8.** What is the maximum number of rectangles that you can count in the given figure?

(a) 2

☐

(b) 3

☐

(c) 4

☐

(d) 6

☐

9. How many edges does a cylinder have?
 (a) 0 ☐ (b) 2 ☐ (c) 3 ☐ (d) 4 ☐
10. Which of the following solids will roll down an inclined plane?
 (a) Cylinder ☐ (b) Cuboid ☐ (c) Cube ☐ (d) Cone ☐
11. Choose the odd one out.
 (a) Circle ☐ (b) Cone ☐ (c) Cylinder ☐ (d) Sphere ☐
12. What is the maximum number of triangles that you can count in the adjacent figure?
 (a) 3 ☐ (b) 4 ☐
 (c) 5 ☐ (d) 6 ☐



QUESTION BAG 2

1. Match the following.

- | | |
|------------------|---------------------------|
| (A) Line | (a) Dice |
| (B) Line segment | (b) No definite length |
| (C) Ray | (c) One curved face |
| (D) Cuboid | (d) Has one end-point |
| (E) Cube | (e) Can be drawn on paper |
| (F) Cylinder | (f) One vertex |
| (G) Cone | (g) Match-box |
| (H) Sphere | (h) 2 circular edges |

2. Name a solid shape.

- (a) with one edge
- (b) with 3 faces
- (c) with one vertex
- (d) with 2 edges
- (e) with one face
- (f) with no vertex



- (g) with no edges
- (h) with only plane/flat faces
- (i) with only curved faces
- (j) with both straight and curved edges
- (k) with both flat and curved faces

3. Answer the following questions.

- (a) Does a cylinder have any straight edge?
- (b) How many vertices does a cylinder have?
- (c) Does a sphere have any edge?
- (d) Can a triangle have all sides different in length?
- (e) Can a rectangle have all sides different in length?
- (f) How many corners are there in a cuboid?

4. Fill in the blanks.

- (a) Two line segments intersect at a
- (b) A has all 4 sides equal.
- (c) The shape of the earth is a
- (d) edges of a cuboid meet at each of its vertices.
- (e) Two adjacent faces of a solid shape meet at a/an
- (f) Two adjacent sides of a plane figure meet at a called the
- (g) A cuboid has pairs of identical faces.

5. State whether each of the following statements is true or false.

- (a) One and only one line can be drawn to pass through two given points
- (b) All the edges of a cuboid are equal
- (c) A cuboid has 8 faces
- (d) A sphere has one face, a cone has 2 faces and a cylinder has 3 faces
- (e) A line segment is the shortest distance between two points on a paper